PRD-FR330V-H1.2 - FR-330V vented front brake

FR-330V vented front brake - PRD (revision H1.2)

> **Fictional, educational design.** Not a real vehicle, product or safety certification. This PRD is the **only** authority for geometry, loads and acceptance in a run. The CTS in `docs/source/` is background: where it and this PRD differ, **this PRD wins** (the differences are listed in section 10).

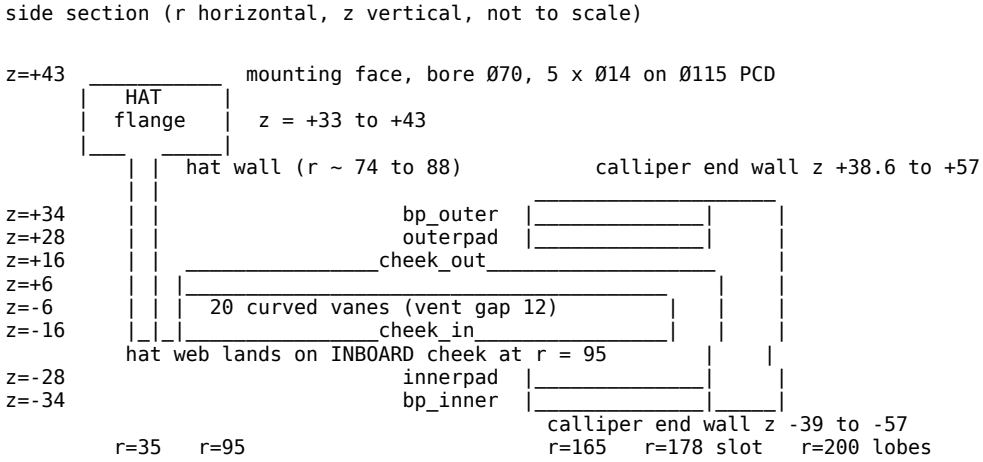
Conventions used everywhere in this document

- Lengths are **millimetres** unless a unit says otherwise. SolidPilot works in **metres**: convert once, at the tool call (`mm / 1000`).
- Ranges are written `a to b` with ASCII minus signs: `z = -16 to -6` means the interval from minus sixteen to minus six.
- **Z is the shaft axis.** Origin at the vent mid-plane. **+Z points outboard** (towards the wheel and the mounting face). Pads are centred on **+X**.
- Angles are **degrees**, measured from +X towards +Y (counter-clockwise viewed from +Z).
- "Checksum" = an independently computed number the built CAD must match. Every checksum in section 4 was computed from the geometry in section 3 by two independent methods that agree within 0.01 %.

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1. What is being built

A vented Ø330 mm grey-iron brake rotor with an integral hat, one pair of pads with steel backing plates, and a scalloped calliper that clears the pad stack. The assembly is **9 part files containing 28 solid bodies**, built in SolidWorks, prepared in SpaceClaim, and solved in Fluent as a transient solid-conduction model of **one** braking stop.



2. Parts, bodies and placement

Build every part **at its true global position** and assemble with **identity transforms**. Nothing is moved after it is built.

#	Part file (exact name)	Bodies	Cross-section	Axial band (z)	Material
1	`cheek_out`	1	annulus r = 95 to 165	+6 to +16	MAT-01 grey iron
2	`cheek_in`	1	annulus r = 95 to 165	-16 to -6	MAT-01 grey iron
3	`vanes`	**20**	curved strips, section 3.2	-6 to +6	MAT-01 grey iron
4	`hat`	1	revolved polygon + 5 bolt holes, section 3.3	-16 to +43	MAT-01 grey iron
5	`outerpad`	1	sector r = 120 to 160, angle -30 to +30	+16 to +28	MAT-02 pad
6	`innerpad`	1	sector r = 120 to 160, angle -30 to +30	-28 to -16	MAT-02 pad
7	`bp_outer`	1	sector r = 115 to 165, angle -30 to +30	+28 to +34	MAT-03 steel
8	`bp_inner`	1	sector r = 115 to 165, angle -30 to +30	-34 to -28	MAT-03 steel
9	`calliper`	1	scalloped body, section 3.4	-57 to +57	(not solved)
	Total	**28**			

Rules that apply to every part:

- **One part file per row.** Only `vanes` holds more than one body. The four rotor parts stay **separate bodies** - never merge cheeks, vanes and hat.
- Touching parts **abut exactly** (shared faces, zero overlap volume). These contacts become the solver's thermal interfaces.
- Build `innerpad` and `bp_inner` **independently** at their own negative z bands. Do not copy-and-edit the outboard parts.

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3. Geometry definitions

3.1 Cheeks

Two flat annuli, r = 95 to 165, each 10 mm thick: `cheek_out` z = +6 to +16, `cheek_in` z = -16 to -6. Both must be **exactly axisymmetric** (no holes, no features): the solver spins them with a solid-motion model that is only valid for axisymmetric bodies.

3.2 Vanes (20 bodies)

Each vane is the region **between two concentric circles** of radius 117 and 123 centred at **C = (20.722, 94.249)**, trimmed radially to **r = 95 to 165**.

- The circle band crosses the rotor annulus on two branches. Keep the branch whose polar angle **increases** with radius: about **-2.3 to +2.3 deg at r = 95**, **+29.8 to +33.4 deg at r = 165**. The other branch (about 153 to 158 deg at r = 95) is wrong.
- Vane k (k = 1 to 20) is vane 1 rotated by **(k - 1) x 18 deg** about Z. Rotate the sketch coordinates, one closed profile per vane. **Do not use a circular-pattern feature** (see the SolidWorks skill: it reports 20 and makes 1).
- Extrude 12 mm **mid-plane** about z = 0 (z = -6 to +6), then trim to the exact r = 95 and r = 165 cylinders so the vane ends are true cylindrical faces that sit conformally against nothing but air.
- Handedness: the vanes sweep towards +theta as r increases. The rotor therefore runs **clockwise viewed from +Z** (negative omega about +Z). See section 5.

3.3 Hat

Revolve this closed polygon, given as (z, r) pairs, 360 deg about Z:

```
(43, 35) -> (43, 86) -> (-6, 88) -> (-6, 95) -> (-16, 95) -> (-16, 78) -> (33, 74) -> (33, 35) -> close
```

Then cut **five Ø14 through-holes** on a **Ø115 PCD** (hole centres at r = 57.5, angles 0, 72, 144, 216, 288 deg) through the mounting flange z = +33 to +43.

- Centre bore Ø70 (r = 35). Mounting face at **z = +43**.
- The web lands on the **inboard** cheek: the hat's cylindrical face r = 95, z = -16 to -6 abuts the bore of `cheek_in`.
- The hat stops at z = -6 below r = 95, so the vent inlet (r = 95, z = -6 to +6) is open. The radial gap between the hat wall (r ~ 88) and the vane roots (r = 95) is about 7 mm.

- SolidPilot revolves with `extrude_feature(feature_type='revolve', ...)`. Verify the hat lands at z = -16 to +43 by measuring. Do not trust a reverse flag.

3.4 Calliper (1 body)

1. **Base** (z = -57 to +57): union of a sector r = 98 to 176, angle -34 to +34, and three R40 circles centred at r = 160, angles -20, 0, +20 deg.
2. **Pad slot** (z = -39 to +38.6): remove sector r = 90 to 178, angle -42 to +42.
3. **Windows** (the **same** z = -39 to +38.6): remove r = 90 to 215 over the angle ranges -42 to -27, -7 to +7 and +27 to +42.
4. **Outer-bridge step**: remove r = 172 to 215 (full circle) for z = +42 to +57 and z = -57 to -42. That is 15 mm in from each axial face.
5. **Piston bores**: six Ø26 blind bores centred at r = 140, angles -20, 0, +20. Three go from the slot face z = +38.6 **outward** to z = +46.6. Three go from z = -39 outward to z = -47. Each is 8 mm deep.
6. No fillets, no chamfers.

The result is **one connected solid**. In the slot band (z = -39 to +38.6) the only material left is two bridges at r > 178, between the windows.

3.5 Pads and backing plates

Sectors centred on +X, angle -30 to +30 deg. Pad friction faces sit **exactly** on the cheek faces (z = +16 and z = -16). Backing plates sit exactly on the pad backs (z = +28 and z = -28).

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4. Checksums - the numbers the CAD must hit

Tolerance: **volume within 1 %**, **centroid within 0.1 mm**, **bbox within 0.05 mm**. If any number misses, the part is wrong: stop, repair it, and re-measure. Never proceed on a miss. SolidPilot reports m^3 and m. Values here are mm^3 and mm.

Body	Volume mm^3	Volume m^3	Centroid (x, y, z) mm	Bounding box / extents mm
cheek_out	571 769.86	5.717699e-4	(0, 0, +11.000)	x,y = -165 to 165; z = +6 to +16
cheek_in	571 769.86	5.717699e-4	(0, 0, -11.000)	x,y = -165 to 165; z = -16 to -6
vane (each)	7 251.63	7.251627e-6	vane_01: (123.842, 40.040, 0)	vane_01: x = 94.922 to 143.121, y = -3.850 to 90.942, z = -6 to +6

vanes (all 20)	145 032.55	1.450325e-4	(0, 0, 0)	each vane centroid at r = 130.154, angle 17.917 + (k-1) x 18 deg
hat (after holes)	513 204.29	5.132043e-4	(0, 0, +18.095)	x,y = -95 to 95; z = -16 to +43
hat (before holes)	520 901.19	5.209012e-4		polygon area 1139.0 mm^2
outerpad	70 371.68	7.037168e-5	(134.600, 0, +22.000)	z = +16 to +28
innerpad	70 371.68	7.037168e-5	(134.600, 0, -22.000)	z = -28 to -16
bp_outer	43 982.30	4.398230e-5	(135.111, 0, +31.000)	z = +28 to +34
bp_inner	43 982.30	4.398230e-5	(135.111, 0, -31.000)	z = -34 to -28
calliper	617 081.48	6.170815e-4	(146.143, 0.000, +0.369)	x = 81.245 to 200.000; y = -98.418 to 98.418; z = -57 to +57
rotor (cheeks + vanes + hat)	1 801 776.57	1.801777e-3		mass at 7200 kg/m^3 = **12.973 kg**

Vane section area is 604.302 mm^2. The older quoted value, 604.2333, is from a polygonal approximation. Both are inside tolerance.

Calliper section-area checks. Use these to localize a calliper miss by band:

Section area mm^2	z band	What is cut in this band
11 891.10	-57 to -47 and +46.6 to +57	outer-bridge step
10 298.32	-47 to -42 and +42 to +46.6	step + piston bores
15 050.62	-42 to -39 and +38.6 to +42	piston bores only
2 310.76	-39 to +38.6	pad slot + windows

Contacts and clearances (checks, not choices)

Pair	Contact type	Expected area / clearance
outerpad / cheek_out, innerpad / cheek_in	face, z = +/-16	5 864.31 mm^2 each
outerpad / bp_outer,	face, z = +/-28	5 864.31 mm^2 each (the

innerpad / bp_inner		pad face)
each vane / cheek_out and / cheek_in	face, z = +/-6	604.30 mm^2 each, 40 contacts
hat / cheek_in	cylinder r = 95, z = -16 to -6	5 969.03 mm^2
hat / vanes	edge only at r = 95, z = -6	**no face contact**
pad stack / calliper slot, axial	34 vs 38.6 and -34 vs -39	4.6 mm outboard, 5.0 mm inboard
pad stack / slot height	slot 77.6 vs stack 68	+9.6 mm (PAD-05 needs >= +4)
rotor OD / slot inner radius of bridges	165 vs 178	13.0 mm
hat wall / calliper end wall	r 86.18 at z = +38.6 vs r = 98	11.8 mm
interference anywhere		**0 mm^3**

Expected thermal surface groups (SpaceClaim uses these to check its work):

Group	Composition	Before imprint mm^2 (SpaceClaim, Gate 2)	After imprint mm^2 (Fluent mesh, Gate 3)
wall-vent	2 cheek inner faces (z = +/-6) + 20 vane flanks and ends	**167 037.60**	**142 865.51** (less 40 vane footprints)
wall-outer	cheek outer faces, OD, cheek_out bore, all exposed hat faces incl. the hat step face z = -6, r = 88 to 95	**240 205.78**	**228 477.17** (less 2 pad footprints)
pad/backing exposed	pad arcs and sides, both backing faces, arcs and sides	**42 997.28**	**31 268.67** (less 2 pad footprints)

SpaceClaim tags whole faces, so a face that is partly a contact is tagged in full (the “before” column). Contact faces that are **entirely** contact (hat r = 95 cylinder, cheek_in bore, pad friction and back faces) are left untagged. Shared topology in Fluent Meshing cuts the footprints out (the “after” column). The hat step face at z = -6 belongs to `wall-outer`, by decision.

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5. Duty cycle - derive it first, show the arithmetic

Inputs: mass **1800 kg**; entry speed **100 km/h**; deceleration **0.8 g** with **g = 9.80665 m/s^2**; front axle bias **0.70**; **two** front rotors; rolling radius **0.33 m**.

Quantity	Formula	Expected value
v0	100 / 3.6	27.7778 m/s
a	0.8 g	7.84532 m/s^2
stop time ts	v0 / a	**3.54068 s**
stopping distance	v0^2 / (2a)	49.176 m
energy per front rotor E	0.5 m v0^2 x 0.70 / 2	**243 055.6 J**
mean power	E / ts	68 646.6 W
peak power (t = 0)	2E / ts	137 293.1 W
wheel speed	v0 / 0.33	84.1751 rad/s (803.8 rpm)
omega0 (signed, about +Z)	clockwise viewed from +Z	**-84.1751 rad/s**
omega_ref for h scaling	(2/3) abs(omega0)	56.1167 rad/s
pad contact area A (each)	(pi/6)(0.160^2 - 0.120^2)	5.864306e-3 m^2
layer heat source, peak	P(0) / (2 A d), d = 0.002 m	5.8529e9 W/m^3
layer heat source, mean	Pmean / (2 A d)	2.9265e9 W/m^3
time steps at dt = 0.01 s	ts / dt	354 full steps **plus** a final 0.000681 s step (ts - 3.54)
sanity: bulk rotor rise if all E stayed in the rotor	E / (m_rotor x cp)	about 41.9 K

Power history: `P(t) = (2E/ts)(1 - t/ts)` for `0 <= t <= ts`, then zero. Speed: `omega(t) = omega0 x max(0, 1 - t/ts)`.

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6. Analysis definition

- **Physics.** Transient heat conduction in **27 solid zones** (the calliper is deleted from the simulation copy). Energy equation on, flow equations off, no fluid domain. Double precision. Initial and ambient temperature 300 K.
- **Heat input.** A frictional heat source in a virtual wall layer of thickness d = 0.002 m, on **one side only** of each of the two pad/cheek contacts. The layer material is **grey iron (MAT-01 values)**; never Fluent's default aluminium. Scale q''' with the closed-form A of section 5, then integrate the applied power with the **measured** mesh contact areas: that integral must equal E within **+/-2 %** (DV-05). The layer is a modelling assumption, not a physical interface.
- **Rotation.** Only `cheek_out` and `cheek_in` get solid motion, omega(t) about +Z through the origin. **Never** rotate the vanes or the hat with this model (they are not axisymmetric). Pre-solve check: the integral of abs((ez x r) . n) dA over the rotating boundaries must be <= 1.592987e-4 m^3/s per rad/s (it is zero for a true axisymmetric cheek).
- **Convection (prescribed inputs, not solved airflow).** Reference h at omega_ref: vent faces 130, outer rotor/hat faces 85, exposed pad/backing faces 40 W/(m^2 K). Scale with speed: `h(t) = max(7, h_ref x (abs(omega(t)) / omega_ref)^0.8)`.

- **Mesh starting point.** Global size 2.2 to 20 mm, growth 1.2, curvature angle 18 deg. Local sizes: cheeks 4 mm, vanes 5 mm, hat 8 mm. Poly-hexcore, no inflation, shared topology, all regions solid.
- **Mesh gates.** 27 solid zones, 0 fluid zones, **45 coupled interface pairs** (2 cheek/pad, 2 pad/backing, 40 cheek/vane, 1 hat/cheek_in). Max **surface** skewness <= 0.90 (volumetric skewness is not reported by Fluent 2026 R1: record it as null with that reason; it does not block), min orthogonal quality >= 0.10, no negative volumes. `wall-vent` meshed area within 2 % of 142 865.51 mm^2. Per-body volume within 2 % of section 4. **Cells + nodes < 512 000** if running Ansys Student.
- **Solve.** dt = 0.01 s through the **full** ts including the final 0.000681 s step. No more than 10 steps between load/speed updates and no more than 25 between h updates; use the exact interval-average of each ramp. Start with 5 iterations per step; the energy residual must reach 1e-8 each step.
- **Energy audit.** (stored + convected) / applied must fall in **90 to 110 %**. Report the residual. Applied / vehicle energy must be within +/-2 %.

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7. Requirements (every one gets a row in the final report)

M = mandatory, S = should, C = could.

ID	Requirement	Target	Class	Evidence expected	Prediction from section 4/5
PKG-01	rotor outer diameter	<= 330 mm	M	CAD measure	330.0
PKG-02	friction ring thickness	32 mm nominal	M	CAD measure	32.0
PKG-03	bolt pattern	5 x Ø14 on Ø115 PCD	M	CAD measure	as built
PKG-04	centre bore	Ø70	M	CAD measure	70.0
PKG-05	mounting face offset	z = +43	M	CAD measure	43.0
PKG-06	wheel rim clearance	>= 6 mm radial	S	package check	**OPEN** - no wheel geometry
MAS-01	rotor mass	<= 13.5 kg	M	volume x 7200	12.973 kg
GEO-01	two cheeks + vanes	present	M	CAD	yes
GEO-02	cheek thickness	10 mm each	S	CAD	10.0

GEO-03	vent gap	12 mm	S	CAD	12.0
GEO-04	curved vanes	>= 18	S	CAD	20
GEO-05	cross-drilled holes	>= 30	C	-	**DEFERRED** to Phase 3: holes break the axisymmetry the solid-motion model needs. Do **not** build them, not even as a display model
GEO-06	hat on inboard cheek, vent inlet open	yes	M	CAD section	yes, 7 mm inlet gap
PAD-01	pad sector angle	60 deg	M	CAD	60
PAD-02	pad radii	120 to 160 mm	M	CAD	yes
PAD-03	pad thickness	12 mm	S	CAD	12
PAD-04	backing plate	6 mm steel, r 115 to 165	S	CAD	yes
PAD-05	calliper clears pad stack	slot >= stack + 4 mm, 0 interference	M	CAD interference	+9.6 mm, 0
MAT-01	rotor grey iron	rho 7200, cp 447, k 52	declared	solver readback	
MAT-02	pad compound	rho 2500, cp 900, k 1.2	declared	solver readback	
MAT-03	backing steel	rho 8030, cp 502, k 16.3	declared	solver readback	
THM-01	rotor friction-surface peak	<= 923 K	M	Fluent	solve
THM-02	pad friction-	<= 623 K	M	Fluent	solve;

	surface peak				**expected to govern**
THM-03	rotor bulk temperature rise	<= 60 K	S	Fluent	about 42 K upper estimate
THM-04	no rotor surface above	1143 K	M	Fluent	solve
THM-05	vent cooling modelled explicitly	distinct vent h	M	Fluent setup readback	yes
THM-06	calliper / brake fluid temperature	<= 428 K	M	-	**DEFERRED**: calliper not in thermal model (DV-09)

Design-verification items: DV-01 package, DV-02 mass, DV-03 inlet open, DV-04 thermal, DV-05 energy within +/-2 %, DV-06 interference - all **in scope**. DV-07 fade/dyno, DV-08 cracking, DV-09 calliper/fluid, DV-10 repeated stops - all **deferred**, with the owner Chassis / Foundation Brake and the target phase Phase 3.

Temperature honesty. Report cell-maximum and wall-layer-maximum temperatures as a range (for THM-04, cell max and wall-facet max over all rotor walls; the layer exists only at the friction faces). If that range straddles a limit, or the wall-normal cell size at the friction face is coarser than the thermal penetration depth sqrt(alpha x t) at the time of the peak, the disposition is **NOT ADJUDICABLE**, not PASS. If the cell maximum alone exceeds the limit, it is **FAIL**. Always record the number of cells across the penetration depth; fewer than 5 is a finding. Lack of a mesh-sensitivity study alone is a finding, not a reason for NOT ADJUDICABLE.

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8. Deliverables

1. `case.json`: the section 5 derivation with every value and unit.
2. SolidWorks: 9 `.SLDPRT`, `brake.SLDASM`, `brake.step`, and `cad_report.json` (each body's measured volume, centroid, bbox and checksum comparison).
3. SpaceClaim: `brake.pmdb` and `tag_report.json` (27 bodies, named selections, areas).
4. Fluent: mesh, case/data, `mesh_report.json`, `setup_report.json`, `results.json` and time-history CSVs.
5. Visuals from **every** stage: CAD isometric and a section through the vent inlet at 9 deg, and the named-selection view, rendered by the spaceclaim-engineer from the live SpaceClaim session's tessellation of the imported STEP (the Geometry API has no window screenshot); mesh images, and temperature contours with units, time and

range. **No 2D drawings in H1.2** (descoped by the user); the drawing is listed as descoped in the report.

6. `report.pdf` with the verdict first: every section 7 requirement with measured value, PASS / FAIL / NOT ADJUDICABLE / OPEN / DEFERRED, evidence key, and the **first change you would make** with its quantified expected effect.

All run artifacts go in a fresh folder `C:\brake-runs\<run-id>\` (never inside OneDrive).

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9. Acceptance of the workflow itself

A run is **complete** only if all 28 bodies passed section 4, the 45 interfaces exist, the energy audit closed, and every requirement has a disposition. The engineering verdict (how many requirements PASS) is reported separately from workflow completion. Otherwise it is **complete_with_findings** or **halted**, and the report says exactly which gate stopped it and with what two numbers (expected vs measured).

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10. Differences from CTS-BRK-0117 rev C

CTS says	This PRD says	Why
GEO-05 >= 30 cross-drilled holes (C)	Deferred, not built	Holes break the axisymmetric solid-motion model. One geometry is built and solved.
hat bore radius 97 mm, 1 mm calliper clearance (older text)	Hat reaches r = 95 at the cheek bore only; hat-to-calliper clearance 11.8 mm	Retired geometry
calliper windows z band not matched to slot	Windows use exactly the slot band z = -39 to +38.6	Removes the sliver faces a mismatched band creates
g = 9.81 in places	g = 9.80665 everywhere	One convention
354 steps x 0.01 s	full ts = 3.54068 s including a final short step	Covers the whole stop
vane intent “pumps under +theta”	vanes as built pump under -theta (clockwise from +Z), omega0 = -84.175	The geometry governs; the intent is reported as unmet